

Workshop Activities

Activity 1 – Types of Analytics

Discuss which type of analytics (descriptive, predictive, or prescriptive analytics) each of the following represents:

1. a data dashboard
2. a model that finds the production schedule that minimises overtime
3. a model that forecasts sales for the next quarter
4. a bar chart
5. a model that allocates your financial investments to achieve your financial goal

Activity 2 – House Price and Square Footage

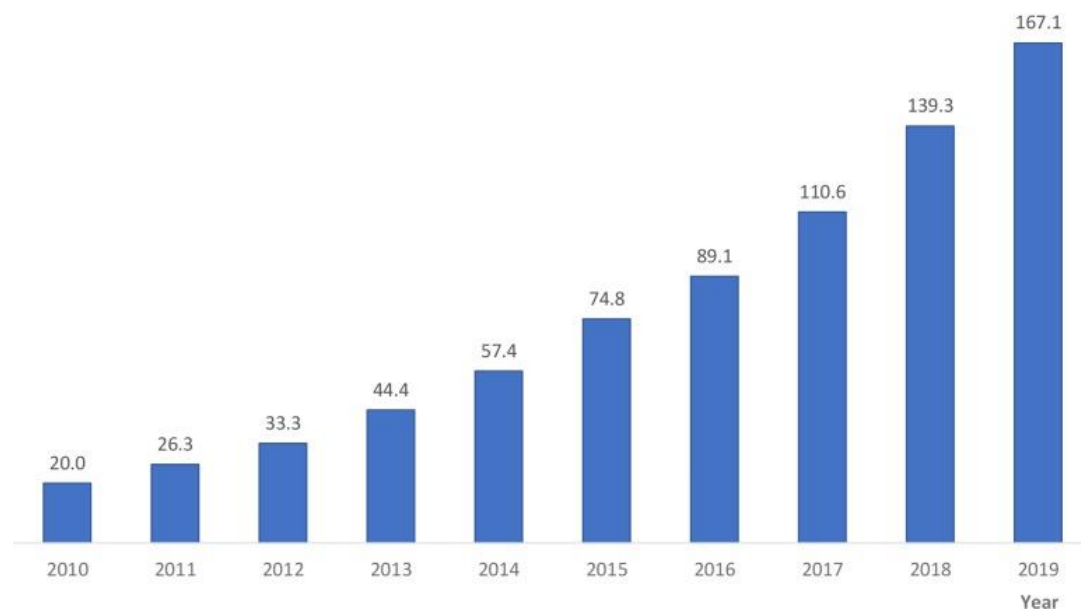
Suppose we want to better understand the relationship between house price and square footage of the house, and we have collected house price and square footage for 75 houses in a particular neighbourhood of Cincinnati, Ohio, from the Zillow website on January 3, 2021.

1. Are these data quantitative or categorical?
2. Are these data cross-sectional or times series?
3. Which of the following type of chart would provide the best display of these data? Explain your answer.
 - Bar chart
 - Column chart
 - Scatter chart

Activity 3 – Netflix Subscribers

The following chart displays the total number of Netflix subscribers from 2010 to 2019.

Netflix Subscribers (millions)



1. Are these data quantitative or categorical?
2. Are these data cross-sectional or time series?
3. What type of chart is this?

Activity 4 – Creating a Chart using Excel

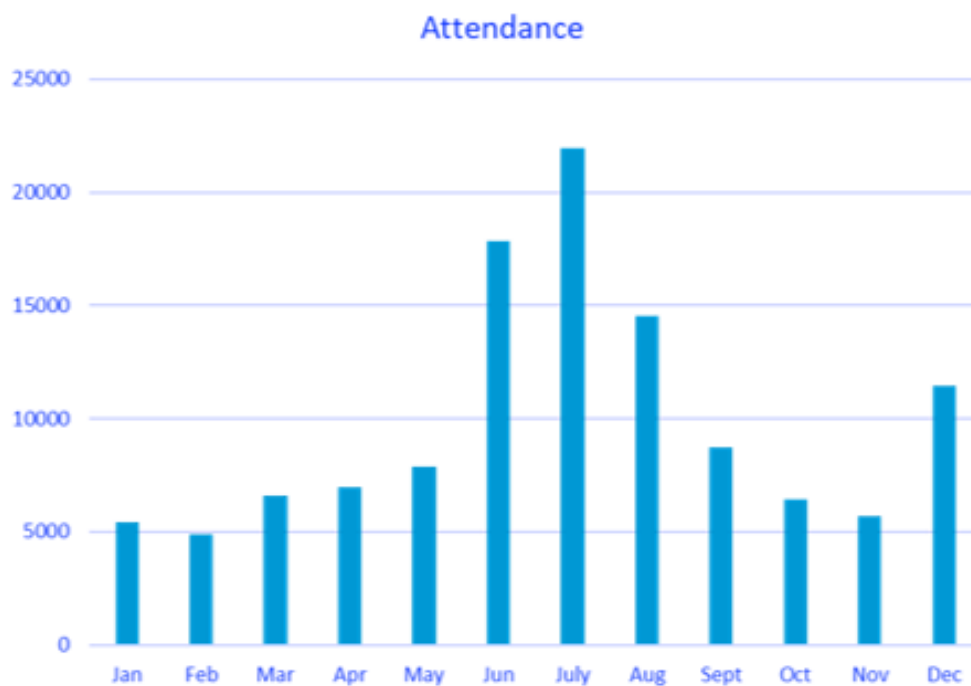
The next activity is to create a chart using data from the Zoo attendance data file.

	A	B
1	Month	Attendance
2	Jan	5422
3	Feb	4878
4	Mar	6586
5	Apr	6943
6	May	7876
7	Jun	17843
8	July	21967
9	Aug	14542
10	Sept	8751
11	Oct	6454
12	Nov	5677
13	Dec	11422

Open the Zoo attendance data file and complete the following steps:

- Step 1. Select cells A1:B13
- Step 2. Click the Insert tab on the Ribbon
- Step 3. Click the Insert Column or Bar Chart button in the Charts group
- Step 4. When the list of column and bar charts subtypes appears, click the Clustered Column button

The column chart you have created should look similar to the following chart:



Activity 5 – Editing a Chart using Excel

The chart created in Activity 4 requires an increased cognitive load. To reduce the cognitive load required to understand the chart, the appearance of it can be improved by making it simpler and better defined. The chart can be refined in various ways using one or more of the following stages. There is no specific order to the stages each refines a different element of the chart.

Stage 1 removes the horizontal grid lines and Stage 2 is how to edit the title.

- Stage 1
 - Click the Chart Elements button

- Deselect the check box for Gridlines and select the check box for Axis Titles
- Stage 2
 - Click the Chart Title text box above the chart and replace “Chart Title” with Zoo Attendance by Month
 - Inside the text box, highlight “Zoo Attendance by Month”
 - Click the Home tab on the Ribbon and in the Font group select Calibri 16
 - To change the position of the title, click on the border of the title box and slide the text box to the left so that the title is above the vertical axis.

Stages 3–5 format the horizontal axis and axis labels.

- Stage 3.
 - Double click any label of the horizontal axis
- Stage 4.
 - When the Format Axis task pane appears, click the Fill & Line button
 - Click Line
 - Select Solid line
 - In the drop down to the right of Colour, under Theme Colours, select Black
- Stage 5.
 - Click the Home tab on the Ribbon and in the Font group select Calibri 10.5

Stages 6–8 format the vertical axis and axis labels.

- Stage 6.
 - Double click any label of the vertical axis
- Stage 7.
 - When the Format Axis task pane appears, click the Fill & Line button
 - Click Line
 - Select Solid line
 - In the drop down to the right of Colour, under Theme Colours, select Black
- Stage 8.
 - In the Format Axis task pane, click the Axis Options button
 - Click Tick Marks
 - Next to Major type, select Inside
- Stage 9.

- Click the Home tab on the Ribbon and in the Font group select Calibri 10.5

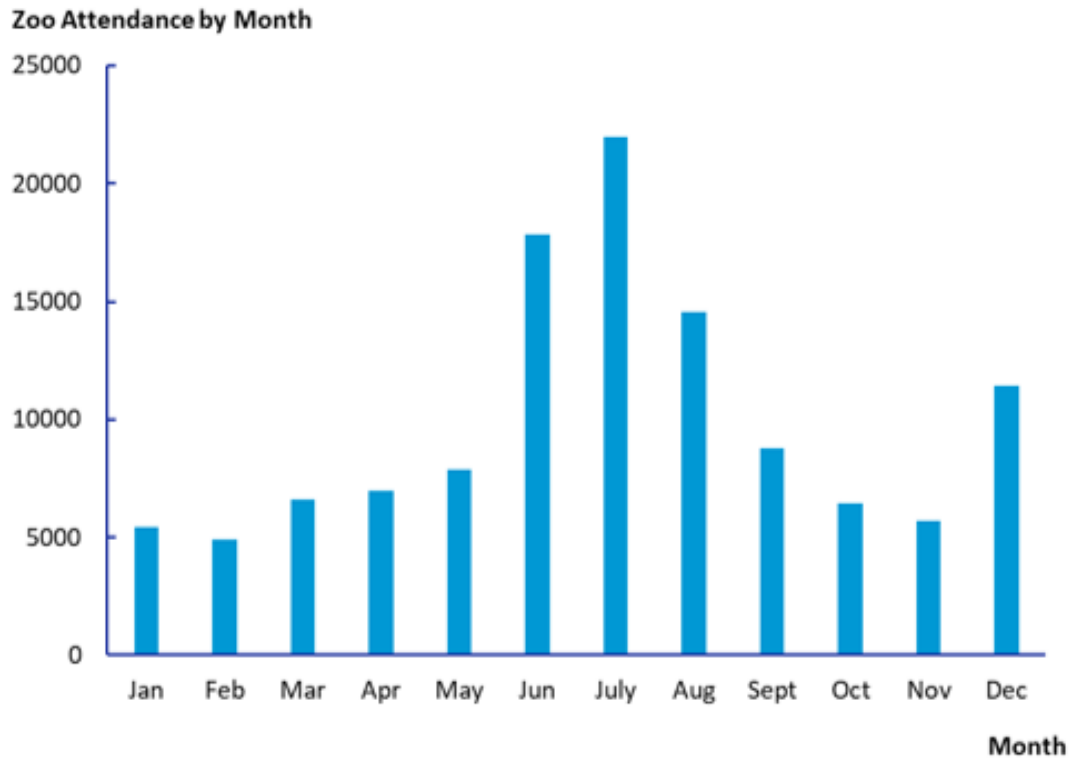
Stages 10–11 add and format axis titles.

- Stage 10.
 - Select the horizontal axis title, place the cursor over the border of the text box and drag it to the right to the end of the axis.
 - In the Font group, select font size 10.5 and click the button.
 - Type Month
- Stage 11.
 - Select the vertical axis title, right click and select Format Axis Title and click the Size & Properties button. Click Alignment and next to Text direction, from the drop-down menu, select Horizontal. Place the cursor over the border and drag it to the top of the vertical axis aligned above the axis labels.
 - In the Font group, select font size 10.5 and click the button.
 - Type Attendance

Stages 12–13 eliminate the border of the chart.

- Stage 12.
 - Click the Chart Area of the chart (anywhere outside of the rectangular area delimited by the horizontal and vertical axes of the chart)
- Stage 13.
 - In the Format Chart Area task pane, click Chart Options
 - Click the Fill & Line button
 - Click Border
 - Click No line

Try to refine the Zoo Attendance chart that you created in Activity 4 to look like the following chart:



The Chart Styles button allows you to change the style of the chart type you have selected, as well the colour scheme of the chart.

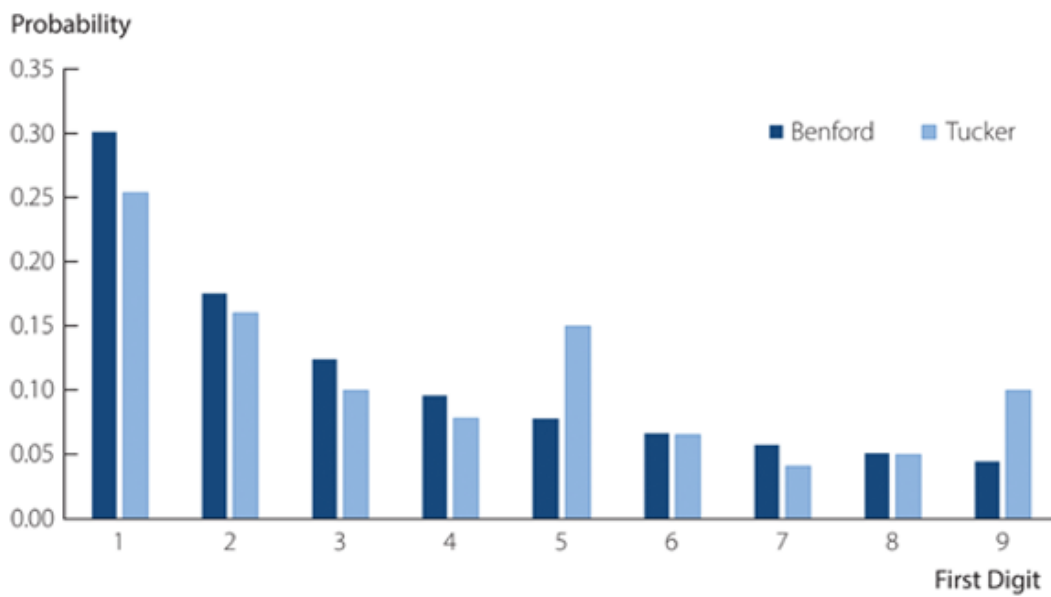
Discuss whether this chart provides a lower cognitive load for the reader/viewer.

Activity 6 – Making Comparisons

Benford's Law, also known as the First-Digit Law, gives the expected probability that the first digit of a reported number takes on the values one through nine, based on many real-life numerical data sets such as company expense accounts.

Benford's Law can be used to detect fraud. If the first digits of numbers in a data set do not conform to Benford's Law, then further investigation of fraud may be warranted. Consider the first digit of Tucker Software's accounts payable (money owed the company) entries in the clustered column chart below.

Benford's Law versus Tucker Software Accounts Payable



1. Does it appear that the data follow Benford's Law? Discuss your answer.
2. Which first digits from Tucker's accounts payable entries stand out as underrepresented in terms of absolute and relative proportional difference for the corresponding expected probabilities as dictated by Benford's Law? Discuss your answer.